

PROJECT DEFINITION

RUF Diamond Parts Portal

Scope, Design, System Architecture, Support and Delivery Plan

Prepared for	Andres Martinez, Service Manager, RUF Diamond Ltd.
Prepared by	Deepshika Ghale
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Currency	All figures in Canadian dollars, excluding tax

This document responds to the requested review areas: hosting, design, system architecture, data structure, functionality, scalability, performance, security, backup, support, ownership and project planning.

1. Infrastructure and Hosting

OVHcloud is recommended. Their pricing is published openly and can be verified on their website at any time, which means the total cost is known in advance with nothing left to discover later.

On data location: OVHcloud hosts in Beauharnois, Quebec, so our data sits in Canada. OVHcloud is a French company, so it is not subject to United States disclosure law. This meets the requirement without paying the premium attached to a Canadian-owned provider.

1.1 The recommended configuration

OVHcloud recommended their VPS-3 for this workload: 6 vCPU, 12 GB RAM, one IPv4 address, up to 2 Gb/s unlimited bandwidth, and 100 GB of storage expandable to 5 TB. This is more than adequate for the portal.

OVHcloud VPS range and current pricing: ovhcloud.com/en-ca/vps

1.2 Total locked-in cost, OVHcloud

Every recurring charge is listed below, including the items that cost nothing. There are no setup fees, no per-user charges, no per-brand or per-model fees, no licence fees, and no data transfer charges on this configuration.

Item	Monthly	Per year	Notes
VPS-3 server	\$16.83	\$201.96	6 vCPU, 12 GB RAM, 100 GB NVMe. Includes IPv4 address, unlimited bandwidth, anti-DDoS protection and standard daily backup.
Premium automated backup	\$1.80	\$21.60	Extends backup retention from 24 hours to 7 rolling days. Recommended.
Snapshot option	\$0.50	\$6.00	Manual restore point taken before significant updates.
Object storage, 500 GB	\$15.00	\$180.00	Drawing images and the independent nightly database copy. Scales with volume.
Domain name	\$0	\$0	A subdomain of rufdiamond.com, which we already own.
SSL/TLS certificate	\$0	\$0	Let's Encrypt, free and renewed automatically.
Database software	\$0	\$0	PostgreSQL. No licence fee, now or as we grow.
Operating system, web server	\$0	\$0	Ubuntu and Nginx. Open source.
Source code hosting	\$0	\$0	GitHub private repository, free at our scale.

Monitoring tools	\$0	\$0	Open source, running on our own server.
Email delivery	\$0	\$0	Free tier covers several thousand messages a month, far beyond our volume.
Setup or onboarding fee	\$0	\$0	None. OVHcloud VPS has no setup charge.
Data transfer charges	\$0	\$0	Unlimited bandwidth included. No egress fees.
Total	\$34.13	\$409.56	Excluding tax. Fixed and predictable.

1.3 Comparison over three years

The ThinkOn quotation is on a three-year term, so the same period is used for both.

	Monthly	One-time	3-year total
OVHcloud	\$34.13	\$0	\$1,228.68
ThinkOn (Quote 302387)	\$474.89	\$465.36	\$17,561.40
Difference			\$16,332.72 lower

What ThinkOn includes that OVHcloud does not. The ThinkOn price covers a managed firewall and managed backup, meaning they configure and watch those for us. On OVHcloud those are set up once and then run automatically: the firewall is standard configuration, backups are scheduled and verified automatically, and both are covered in Sections 9 and 10. This is the practical difference between the two figures.

1.4 What is not in this figure

So that there is no ambiguity, the total above covers infrastructure and third-party services only. It does not include:

- Development of the portal itself.
- Ongoing technical support, meaning whoever patches the server, watches the backups, and responds if something breaks.

Both sit outside this section. Everything that OVHcloud and the supporting services will charge us is in the table in 1.2.

1.5 Cost as the catalog grows

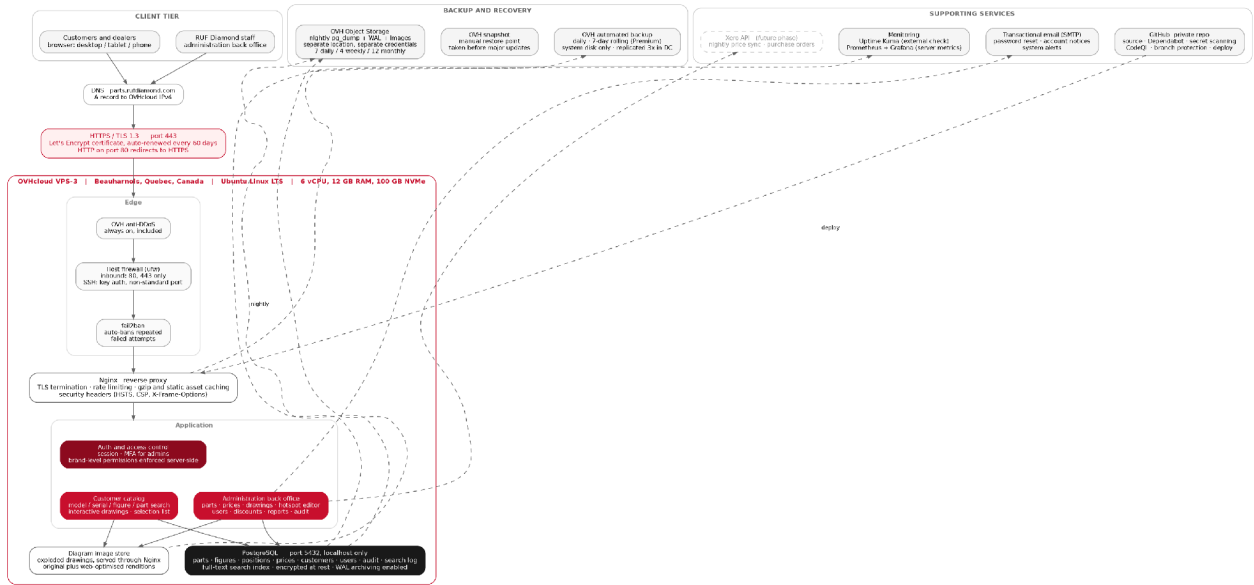
The figure above does not change as we add machines or brands. There are no per-model or per-brand fees, and no per-user charges, which is a structural difference from the platform we use today. The only cost that grows is storage, and only if the drawing library becomes large.

If we	Cost change	Notes
Add the IronHorse line	\$0	No brand fee. The structure already supports three brands.
Add more machines	\$0	No per-model fee.
Add more customer accounts	\$0	No per-user fee.
Exceed 500 GB of storage	Approx. \$0.03 per GB per month	Only if the drawing library grows substantially.
Outgrow the server	Approx. \$15 more per month	The next VPS size up. Not expected within the pilot or the first phase.

A saving plan on a longer commitment would reduce the server cost further. The figures above use the standard month-to-month rate, so they represent the upper end rather than the best case.

2. System Architecture

The portal is a single self-contained system running on one server in Quebec, on its own web address, completely separate from the main RUF Diamond website. Neither can affect the other.



System architecture. Everything inside the red box runs on the OVHcloud server in Beauharnois, Quebec.

2.1 In plain terms

A customer opens parts.rufdiamond.com in a normal web browser. The connection is encrypted, the same way online banking is. The server checks who they are and which product lines they are allowed to see, then shows them the catalog for those machines only.

Behind that, three things do the work. A database holds all the parts, drawings, prices and customer records. The portal itself is the software customers and our staff actually use. And a set of automatic

routines run in the background: backups every night, security updates, and a check every couple of minutes confirming the site is still up.

Everything sits on one server because a catalog used by a few dozen people does not need more. If it ever does, moving to a larger server is a straightforward change, not a rebuild.

2.2 The main parts

Part	What it does
The portal	What customers see: search, exploded drawings, parts lists, and the request list. Also the administration side that our staff use to update parts, prices, drawings, customers and discounts.
The database	Holds everything: parts, drawings, prices, customers, users, and a record of every change made. It is reachable only by the portal, never directly from the internet.
Drawing storage	The exploded diagram images, stored so they load quickly and copied nightly to a separate location.
Protection	Encrypted connections on every page, a firewall accepting only web traffic, automatic blocking of repeated failed logins, and OVHcloud's own protection against traffic floods.
Backup	OVHcloud backs up the server daily. On top of that, the database and drawings are copied every night to storage in a separate location. Covered in Section 10.
Monitoring	An automatic check from outside confirming the portal is reachable, with an alert if it is not.
Source code	Held in a private GitHub account owned by RUF Diamond. The portal is updated from there rather than edited directly on the server.
Xero (later)	Shown dashed on the diagram. A future addition, not part of the initial build. Section 6.

2.3 What it is built with

All of it is free, open-source software. That is deliberate: no licence fees now or later, and no dependence on any one supplier's product.

Element	Choice	Why, and link
Server	OVHcloud VPS-3	Published pricing, Canadian data centre in Quebec. www.ovhcloud.com/en-ca/vps/
Operating system	Ubuntu Linux LTS	Security updates for five years. The most widely used server platform, so help is easy to find. ubuntu.com
Database	PostgreSQL	No licence fee. Handles the catalog and the search without needing a separate product. www.postgresql.org

Web server	Nginx	Free, and efficient at serving large drawing images. nginx.org
Security certificate	Let's Encrypt	Free and renews itself, so it cannot lapse by oversight. letsencrypt.org
Source code	GitHub	Private repository under a RUF Diamond account, with free security scanning. github.com

2.4 Two copies of the portal

Two environments are recommended, which is the normal minimum. The live one is what customers use. A second, identical copy is used to test changes before they go anywhere near customers. It runs on the same server at no extra cost. Testing an update on a copy first is what prevents most avoidable outages.

2.5 How updates reach the portal

Changes are published from GitHub rather than edited directly on the live server. This means every change is recorded and attributable, any release can be rolled back to a previous version, and nobody has to remember what was altered by hand.

3. Portal Design and User Experience

Design status

The portal design has been prepared in Figma. It covers the customer catalog and the administration back office, with responsive layouts for desktop, tablet and mobile, and an interactive prototype showing navigation, search and drawing interaction.

Figma design file:

[\[insert Figma share link here\]](#)

The Figma file serves as the visual and interaction specification for development. Comments can be left directly on the designs for review.

Screens designed

Fourteen screens cover the full customer journey and the administration side.

	Screen	Purpose
1	Sign in	Customer and staff login, with password recovery.
2	Select machine	Choose the machine, filtered to the product lines the account is permitted to see.
3	Serial range	Choose the serial-number variant, so the parts shown match the machine in front of the customer.
4	Find a part	Direct search by part number or description, for customers who already know what they need.

5	Systems	The assembly groups: cabin, engine, hydraulic and the rest.
6	Figures	The drawings within a system, shown as thumbnails.
7	Figure drawing	The exploded drawing with its parts list. Selecting a part highlights every place it appears.
8	Request list	The parts the customer has selected, with quantities, before submitting.
9	Confirmation	Confirmation that the request has been sent, with a reference.
10	Dealer figure net pricing	The same drawing view for a dealer account, showing net pricing after their discount.
11	Dealer order summary	Dealer request summary with net pricing and totals.
12	Admin view for figures	The administration side: upload a drawing, place the callouts, link them to parts.
13	Search with no results	What a customer sees when nothing matches, with suggested next steps. Also logged so that catalog gaps surface in reporting.
14	Empty request list	The request list before anything has been added.

The last two matter more than they appear to. Designing the empty and no-result states deliberately is what stops a portal feeling broken when a customer hits one.

Design deliverables

Deliverable	Contents
Structure and user flows	Site map, key user journeys, search paths and administration workflows.
Wireframes	Layouts for desktop, tablet and mobile.
Visual design	High-fidelity customer portal and administration screens.
Interactive prototype	Clickable prototype covering navigation, search, drawing interaction and key administration tasks.
Handoff package	Component library, typography, spacing, responsive states, icons and assets, and implementation notes for the developer.

Customer experience

The design retains the methods customers already use, and improves the points that currently cause friction.

- Find parts by machine model, serial number, machine image, part number or description.
- Exploded diagrams displayed beside the corresponding parts list.
- Every occurrence of a selected part highlighted on the drawing, not only the first.
- Diagram and list selection synchronized in both directions.

- Search that tolerates partial terms and common spelling errors.
- Zoom, print, email and shopping-selection functions.
- Responsive layouts for desktop, tablet and mobile.
- The most recent update date shown for parts and pricing.

Responsibility for design, build and future changes

Activity	Responsibility
Design	Complete. The Figma file covers the customer catalog and the administration back office, and serves as the specification for development.
Build	A development resource working to the approved Figma design and the requirements in this document. The technical build, database work and hosting configuration should be assigned explicitly rather than assumed.
Content changes after launch	RUF Diamond, through the administration back office. Text, prices, parts, drawings and customer settings need no technical involvement.
Interface and structural changes	Designed and approved first, then implemented by the development resource. Screen layout and workflow changes are not made directly against the live system.
Future enhancements	Raised, prioritised and scheduled as described in Section 11. New capabilities follow the same route: design, approval, build, test, release.

Change management

Major screen and workflow decisions are approved in Figma before development begins. Content changes are made afterwards through the administration portal. Structural interface changes should be designed and approved before implementation.

4. Database Structure

The database is the system of record and is owned entirely by RUF Diamond. It is structured to mirror how the equipment is actually organised, which is what allows new brands and models to be added without rebuilding anything.

4.1 Catalog structure

The catalog is held in seven linked tables. The table below is a high-level summary of what each one holds.

Table	Purpose
BRAND	Fat Truck, IronHorse, Agilis. Adding a brand is a new row, not a rebuild.
MODEL	The machine, for example FT3 Wagon or 2.8C.
VARIANT	Serial-number ranges within a model, for example 2.8C up to 9928C210502 and 9928C210503 and up.

ASSEMBLY	The system groupings: cabin, drive, electric, engine, filters, frame, fuel, hydraulic, tire and wheel.
FIGURE	An exploded diagram belonging to an assembly, with its image file.
POSITION	One row for every callout on a drawing, holding the reference number, quantity and the coordinates where it sits. A part appearing three times on a figure has three rows.
PART	The part itself, stored once and reused by every figure and model that references it.
PRICE	Held separately with an effective date, so pricing history is retained rather than overwritten. Also carries the field reserved for a future Xero item code.

Why POSITION is a separate table. This is what solves the highlighting issue raised in the review. Because each callout is its own record rather than one record per part, selecting a part highlights every location where it appears, the quantity in the list and the number of highlighted points always agree, and clicking any point on the drawing selects the correct row in the list.

4.2 Customers, access and audit

A second group of tables covers who may see what, at which price, and what they did.

Table	Purpose
CUSTOMER_CATEGORY	Dealer, Retail and any category added later. Carries the discount percentage, currently 10% and 0%.
CUSTOMER	The company. Holds an optional discount override for individually negotiated accounts, and whether the account may submit orders.
CUSTOMER_BRAND_ACCESS	Which product lines a customer can see. A Fat Truck customer has one row, a dealer carrying two lines has two.
USER	Individual login accounts belonging to a customer, or to RUF Diamond staff.
AUDIT_LOG	Every change recorded with the user, the record, the old and new values and the timestamp.

Discounts are calculated when a page is displayed rather than stored against each part. Changing the dealer discount from 10% to 12% is a single field edit that takes effect immediately across every part and every machine.

4.3 How the FT3 Wagon export maps to the database

The FT3 Wagon export dated 14 July 2026 contains 635 rows across 16 columns. Every column has a defined destination in the new structure, and the mapping below is what the import tool will implement.

Export column	Destination	Notes
No.	not stored	Row sequence in the export only. Ordering is held in the new structure by sort_order.

PART NO	PART.part_no	536 unique values across 635 rows. Becomes the unique key for a part.
DESCRIPTION	PART.description	505 unique values. Indexed for search.
MODEL	BRAND.name + MODEL.name	Currently a single combined value, FAT TRUCK FT3 WAGON. Split on import into brand and model so IronHorse and Agilis sit alongside it.
VARIANTS	VARIANT.label, serial_from, serial_to	SERIAL NUMBER 99FT3WXXXXXX and up. Parsed into a from and an optional to value.
AGGREGATE / SYSTEM	ASSEMBLY.name	12 values on this machine: CABIN, ENGINE, HYDRAULIC, FILTERS, DRIVE SYSTEM and so on.
GROUPNO	FIGURE.figure_no	44 distinct figures, for example FT3 WAGON FIG- 1.1.
ASSEMBLY NAME - PAGE	FIGURE.name	The readable name of the drawing.
START DATE	PRICE.effective_from	Currently 2026-01-01 throughout. Becomes the start of the price record.
END DATE	PRICE.effective_to	Empty in the export, meaning the price is current. Held as null.
QTY	POSITION.quantity	Quantity used at that location on that drawing.
PNC	POSITION.ref_no	The callout number printed on the drawing. This is what the reference arrow points to.
UNIT PRICE (CAD)	PRICE.amount, PRICE.currency	262 distinct prices. Currency recorded explicitly as CAD.
MAKER	PART.manufacturer	Empty in the current export. The field is provided so it can be populated later without a structural change.
S/NS	PART.stock_status	Currently S on every row.
REMARKS	POSITION.remarks	148 rows carry a remark, for example TOW PACKAGE OPTION or MIDDLE NOT SHOWN. Held against the position rather than the part, since the note applies to that location on that drawing.

The one thing the export cannot carry. There is no column for where on the drawing each callout sits. The coordinates are the only data that must be created rather than imported. This is done once per figure in the hotspot editor, by opening the drawing and clicking each callout. Because the parts and reference numbers are already loaded from the export, this is clicking rather than typing, and it is work RUF Diamond staff can do without technical knowledge.

What the numbers tell us. 635 rows against 536 unique part numbers means 83 parts appear on more than one figure of this single machine, one of them on five separate drawings. Under the current system each of those has one callout recorded and the rest are unmarked. This is precisely the behaviour raised in the review, and storing each callout as its own row is what corrects it.

4.4 Data migration

The FT3 Wagon Excel export is the pilot dataset. The sequence for each machine is the same:

1. Export the parts data from the current system.
2. Clean and map the data to the new structure.
3. Import parts, descriptions, quantities, prices and remarks.
4. Upload the associated drawings.
5. Mark the callout positions on each drawing to link them to the parts records.
6. Validate with RUF Diamond before release.

Steps one to four are quick and largely automatic once the import tools exist. Step five is the manual work, since a spreadsheet cannot record where on a drawing each number sits. The pilot measures how long this takes for one machine, which then makes the remaining machines predictable.

Outstanding input: the FT3 Wagon drawing files. Vector or high-resolution files are preferable, although the best available export can be used for the pilot.

4.5 Weekly price updates

Pricing is received weekly, so the update route needs to be quick and low-risk rather than a data-entry exercise. Two routes are provided.

Route	How it works
Bulk spreadsheet update	Export the current price list to Excel, edit the price column, upload it back. The portal matches on part number, shows what will change before anything is applied, and asks for confirmation. A weekly price file becomes a few minutes of work rather than a day of typing.
Individual edit	Change a single price directly in the administration screen. Takes effect immediately.

In both cases the previous price is retained rather than overwritten, so we can always see what a part cost at a given date. Every change is recorded against the person who made it. Changes are live to customers as soon as they are confirmed, so the catalog stops being a week behind.

Once the Xero integration in Section 6 is in place, the weekly file can flow in automatically rather than being uploaded, at which point the update becomes no work at all.

4.6 Ongoing maintenance by RUF Diamond

All of the following are done through the administration portal, with no technical assistance required:

- Add, edit, archive or remove parts.
- Update prices individually or through bulk spreadsheet import.

- Edit descriptions, quantities, remarks and related information.
- Upload or replace diagrams and mark part locations.
- Add manufacturers, product lines, machine models and serial-number variants.
- Manage customer accounts, product visibility and discounts.

5. Portal Functionality and Administration

Customer portal

- Secure login and account recovery.
- Machine and parts search using multiple entry points.
- Interactive exploded drawings with synchronized parts lists.
- Part detail showing description, quantity, remarks, price and update date.
- Shopping selection or enquiry list, with ordering to follow if approved.
- Print, email and export functions.
- Responsive use on desktop, tablet and mobile.

Administration portal

- Parts, prices, descriptions, quantities and remarks management.
- Spreadsheet-based bulk imports and updates.
- Diagram upload, replacement and part-location mapping.
- Manufacturer, product-line, model and serial-range management.
- Customer account creation, disabling, password reset and access control.
- Customer categories, discount rules and product visibility.
- Audit history showing who changed what and when.
- Report viewing and export to Excel.

Reporting

Report	Purpose
Most-searched parts	Identifies demand and frequently accessed items.
Searches with no results	Highlights missing descriptions, synonyms or catalog gaps.
Machine and section activity	Shows which product areas are used most.
Customer activity	Shows login frequency, last access and portal usage.
Viewed but not selected	Supports future sales and catalog improvement analysis.
Change history	Provides accountability for catalog and pricing changes.
Usage trends	Tracks visitor volumes and activity over time.

6. Xero Integration

Linking the portal to Xero is a stated priority. It is achievable, and this section sets out what it would do, how it would work, and what is being built now so that it is straightforward later rather than a rebuild.

6.1 The problem it solves

Pricing arrives weekly. Without integration, that price is entered in Xero for accounting and again in the portal for customers. Two places, entered twice, and they drift apart the moment someone forgets one of them. A customer then sees a price that no longer matches what we would invoice.

With integration, the price is entered once and the portal follows automatically.

6.2 A worked example

Take a real part from the FT3 Wagon catalog. Part 36-00304, the air filter element, currently listed at \$279.70.

Step	What happens	Detail
1	The price changes in Xero	Our supplier raises the cost, and 36-00304 is updated to \$291.40 in Xero as part of the normal weekly accounting work. Nobody touches the portal.
2	Overnight, the portal asks Xero what has changed	A scheduled routine runs at, say, 2am. It requests from Xero only the items modified since it last ran, rather than the whole catalog.
3	The portal matches the item	Xero returns item code 36-00304. The portal finds the part holding that same code in its Xero reference field and knows they are the same item.
4	A new price record is added	The portal adds a new price of \$291.40 effective from today. The old \$279.70 record is closed off, not deleted, so we can still see what the part cost last month.
5	Customers see the new price	A retail customer opening the drawing sees \$291.40. A dealer on 10% sees \$262.26, calculated at the moment the page is drawn.
6	The change is logged	The audit trail records that the price changed, when, and that it came from the Xero sync rather than a person.

Total human effort in that example: entering the price in Xero once, which is work already being done.

6.3 Later: requests becoming purchase orders

The same connection can work in the other direction. When a customer submits a parts request, the portal can create a draft purchase order or invoice in Xero with the correct item codes, quantities and prices already filled in, ready for review. The alternative is re-keying it, which is both slower and where transcription errors come from.

This is a second stage, sequenced after price synchronisation is working reliably.

6.4 What is built now, at no additional cost

Three decisions are made at the outset. All three cost nothing now and are expensive to retrofit later.

Decision	Why it matters
A Xero item code field on every part	Sits empty until we integrate. Without it, parts would later have to be matched to Xero by description, which is guesswork. With it, matching is exact.
Prices held with effective dates	Lets a sync add a new price without erasing the old one, so pricing history survives. Already required for our own purposes, and it makes the integration simpler.
All pricing read through one place in the code	When pricing later comes from Xero, one component changes rather than every screen that shows a price.

6.5 How it connects

Xero publishes a documented interface for exactly this purpose, covering items with purchase and sales prices, contacts, invoices and purchase orders.

Xero developer documentation: developer.xero.com

Two practical points. The connection is authorised once by a RUF Diamond Xero user and then runs unattended. And it runs on a schedule overnight rather than asking Xero every time a customer opens a page, because Xero limits how often it can be called and because the portal must keep working normally if Xero is briefly unavailable.

6.6 Sequencing

Xero integration is a later phase, not part of the initial build. Until it exists, prices are maintained in the portal through the weekly spreadsheet upload described in Section 4.5, which takes minutes. After it, Xero becomes the source and the portal follows. Either way there is one place prices are maintained, never two.

7. Users, Capacity and Scalability

Planning question	Design assumption
Registered users	Several thousand accounts are supportable. The expected operating range is in the low hundreds.
Concurrent users	The environment should be designed and load-tested for at least 50 to 100 simultaneous users, well beyond expected demand.
Geographic access	Accessible internationally. Users far from the Quebec data centre may see slightly higher page load times.
Storage growth	Drawings are the main storage driver. The VPS-3 includes 100 GB and expands to 5 TB without changing the application.

Scaling path	Increase the server size first, separate the database when justified, and add further application instances only if demand requires it.
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Capacity targets should be confirmed by load testing before launch. The architecture allows growth without changing the user experience or rebuilding the data model.

8. Performance and Monitoring

Measure	Target	Validation
Typical page load	Under 2 seconds	Measured under agreed test conditions.
Search results	Under 1 second	For part-number and description searches.
Large drawing load	Under 3 seconds	Images optimized without making labels unreadable.
Monthly availability	Target 99.9%	Subject to hosting and support arrangements.
Application error rate	Below 0.1% of requests	Exceptions logged, reviewed and corrected.

Monitoring tools

Monitoring can be built entirely from established open-source tools at no licence cost. All of the following are free and widely used.

Tool	Purpose	Link
Uptime Kuma	Checks from outside the server that the portal is reachable, every minute or two, and alerts immediately if it is not. Covers the most important case.	uptime.kuma.pet
Prometheus	Collects server and database measurements over time.	prometheus.io
Grafana	Displays those measurements as dashboards, with alerting.	grafana.com
Netdata	A lighter alternative to the two above. Installs in minutes and gives live server metrics out of the box.	www.netdata.cloud
GoAccess	Summarises web traffic and error patterns from the server logs.	goaccess.io
Backup verification	Automated confirmation that each nightly backup completed, with an alert when one does not. Backups fail silently more often than they fail loudly.	Scripted, no product required.

A monthly summary covering availability, performance, incidents and capacity should be produced from these tools and reviewed.

9. Cybersecurity

Expected threats

Most of what a portal like this faces is automated rather than targeted.

- Automated scanning for outdated or vulnerable software components.
- Repeated login attempts using passwords stolen from other websites.
- Attempts to attack through forms, search fields or file uploads.
- Compromise of an administrator account.
- Traffic floods intended to make the site unavailable.
- **Accidental damage from our own side**, such as an incorrect bulk update or a deletion. In practice this is the most likely cause of a bad day, and also the easiest to recover from.

Standard protections

These are the ordinary measures expected of any business web application. None of them is exotic or expensive.

Control	Detail
HTTPS and TLS	All pages encrypted. Certificate issued free through Let's Encrypt and renewed automatically, so it cannot lapse by oversight.
Two-step login for administrators	A code from a phone app in addition to the password. This single measure prevents most account takeovers.
Role-based permissions	Each account reaches only what it should, enforced by the application rather than by hiding menu items.
Password and lockout rules	Minimum password standards, and automatic blocking after repeated failed attempts.
Secure coding practices	Input validation and safe handling of database queries and file uploads, which prevents the common attack types.
Patching	Automatic security updates for the operating system, and scheduled updates for the application components.
Firewall and rate limiting	Only web traffic accepted. Administrative access restricted by key. Repeated abusive requests blocked automatically.
Encrypted database and backups	Data encrypted at rest, and backups held where they cannot be deleted through the portal itself.
Audit log	Every administrative and data change recorded, so we can always establish what happened and who did it.

Automated code and dependency scanning through GitHub

Because the source code is held in GitHub, several security tools are available at no additional cost and can be enabled from the outset. These run automatically rather than depending on anyone remembering to check.

GitHub feature	What it does
Dependabot alerts	Watches the third-party components the portal depends on and raises an alert when a security flaw is published for one of them. This is the single most common way web applications are compromised.
Dependabot updates	Automatically prepares the update for a vulnerable component so it can be reviewed and applied quickly.
Secret scanning	Detects passwords, keys or connection details accidentally committed to the repository and alerts before they can be misused.
Push protection	Blocks a commit containing credentials before it reaches the repository at all.
CodeQL code scanning	Analyses the portal's own code for common security weaknesses each time a change is made, and reports them for correction.
Branch protection	Prevents changes being made directly to the released version without review.

These features are free for private repositories and should be switched on when the repository is created, not later. Details: docs.github.com/en/code-security

Incident response

The response sequence is: contain the incident, establish what was reached and when, reset affected credentials, restore from a known-good backup taken before the incident, close the gap that allowed it, document what happened, and notify anyone whose information was involved where required. Privacy law requires notification where personal data is affected, and customer contact details count.

The party responsible for responding should be named before the portal goes live rather than identified during an incident.

10. Backup and Recovery

OVHcloud provides backup facilities directly, and these form the first layer. A second, independent layer is added because of a specific limitation described below.

10.1 What OVHcloud provides

Facility	Detail
Standard automated backup	Included free with every VPS. A full backup of the system disk taken daily and retained for 24 hours. Restored from the OVHcloud control panel without needing to connect to the server.
Premium automated backup	Paid option. Extends retention to seven rolling days and allows the backup time to be set, so it can run outside business hours. Recommended.

Replication	Backups are copied three times within the data centre, on servers separate from the one hosting the portal.
Snapshots	A manual point-in-time image of the server, taken before a significant update so it can be rolled back. One snapshot is held at a time.
Retention after cancellation	Data is archived for 30 days after a VPS is cancelled, then deleted.

An important limitation to be aware of. OVHcloud's automated backup and snapshots cover the system disk only. Any additional storage volume attached to the server is not included. If the database or the diagram files were held on an attached disk, they would not be in the OVHcloud backup. This is a documented behaviour and a known cause of unpleasant surprises, so the design must account for it deliberately rather than assume the provider backup covers everything.

10.2 Our backup design

Two layers, so that no single failure loses the catalog.

Item	Frequency	Purpose
OVHcloud automated backup	Daily, 7-day retention	Full server recovery. Covers the operating system and everything on the system disk.
Database dump to object storage	Nightly	An independent copy of the catalog data, written to storage separate from the server. This is the layer that protects against the system-disk limitation above.
Database transaction log	Continuous	Allows recovery to a chosen minute rather than only to last night, which matters after an incorrect bulk update.
Diagram image files	On change, versioned	Every version of every drawing retained, so replacing one by mistake is reversible.
Server snapshot	Before major updates	A rollback point taken manually before any significant change.

10.3 Where backups are held and for how long

- Three copies across two systems, with one copy in a location separate from the production server.
- Object storage credentials kept separate from the portal's own credentials, so that access to the portal does not grant access to the backups.
- Retention of seven daily, four weekly and twelve monthly recovery points. The monthly copies cover the case of an error found long after it was made.
- Recovery procedures documented and stored outside the system they describe, so they are readable when the system is unavailable.
- A recovery test carried out and recorded quarterly. A backup that has never been restored is an assumption rather than a backup.

10.4 How RUF Diamond would reach the backups

Server backups are restored through the OVHcloud control panel, using our own account credentials. Day-to-day corrections do not require a restore at all: because every change is recorded in the audit log, an incorrect price or description is simply corrected in the administration screen. The independent database copies in object storage are reachable through our own account and can be downloaded directly if ever needed.

10.5 Recovery times

Scenario	Target	Approach
Incorrect price or description	Immediate	Corrected in the administration portal. No restore required.
Incorrect bulk update or deletion	1 to 2 hours	Restore to a verified point before the change, confirm, then bring into production.
Server or data-centre failure	2 to 4 hours	Rebuild the server, restore the database and reconnect the image storage.

Objective: no more than 15 minutes of work lost in any failure, and the portal running again within four hours in the worst case.

11. Support and Maintenance

Level	Responsibility	Activities
Content administration	RUF Diamond	Prices, descriptions, parts, diagrams, customer accounts, discounts and routine catalog changes.
Technical operations	Appointed technical resource	Server and application updates, security patches, monitoring, certificate checks, backup verification and incident response.
Changes and additions	Appointed technical resource	New features, workflow changes and future integrations.

The content side sits with RUF Diamond and requires no technical background. The technical side is a distinct responsibility and should be assigned explicitly rather than assumed, since it includes availability outside working hours.

12. Ownership, Source Code and Documentation

Everything produced for this project belongs to RUF Diamond. The points below should be established at the outset rather than settled later.

Item	Position
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Source code	Owned by RUF Diamond.
Database and content	Owned entirely by RUF Diamond, exportable in common formats at any time and at no cost.
Code repository	A private GitHub repository under a RUF Diamond account. See below.
Figma design files	The Figma project, component library and assets belong to RUF Diamond.
Technical documentation	Architecture, data model and operational notes, sufficient for another qualified developer to continue the work.
Administrator guide	Task-based instructions for RUF Diamond staff.
Deployment documentation	Instructions for configuring and rebuilding the system on the hosting environment.

GitHub repository

The source code will be held in a private GitHub repository. RUF Diamond can create the organization account at any time, and doing so early is worthwhile because it means all work is held in our own account from the first line of code rather than moved across later.

Item	Position
Account	A GitHub organization created under RUF Diamond, with RUF Diamond holding the administrator rights.
Cost	Free. Private repositories and the security scanning features described in Section 8 are included at no charge at our scale.
Visibility	Private. The source code is not publicly accessible.
Access	Anyone working on the portal is added as a collaborator and removed when their involvement ends.
History	The full development history stays with RUF Diamond, giving a complete record of what changed and when.
Documentation	Technical and deployment documentation stored alongside the code so it stays current with it.
Credentials	Passwords, keys and connection details are never stored in the repository.
Repository backup	A periodic copy held outside GitHub, so the code remains available if the account or service is unavailable.

At least two RUF Diamond administrators should hold access to the GitHub organization and to the OVHcloud account, so that neither depends on a single individual.

Creating the accounts. Both the GitHub organization and the OVHcloud account should be opened using a RUF Diamond email address rather than a personal one, so that ownership sits with the company from

the start and does not need transferring later. This can be done now, before any development begins, and costs nothing in the case of GitHub.

Keeping the system maintainable

- Widely adopted, well supported technologies only, with no proprietary components that tie us to one supplier.
- Documentation updated as features are delivered rather than promised at the end.
- Independent RUF Diamond access to hosting, the GitHub repository, Figma files, backups and administrator credentials.
- An annual technical and security review.

13. Project Phases and Delivery Plan

The design work is already done, which removes the phase that usually takes longest to settle. Development is being carried out using AI-assisted tooling, which substantially shortens the build compared with conventional estimates. On that basis the pilot should be complete in roughly two to three months rather than the six to nine a project of this description would normally take.

No development costs are included here.

13.1 Phases

Phase	Scope and deliverable	Duration	Status
1. Design	Screens, user flows and interactive prototype in Figma, covering the customer catalog and the administration side. Fourteen screens, listed in Section 3.		Provided
2. Requirements sign-off	Confirm the design and this document reflect what is wanted, before building starts.	1 week	Pending review
3. Accounts and environment	Open the OVHcloud and GitHub accounts, provision the server, apply security settings, connect the subdomain and certificate.	Under 1 week	Ready to start
4. Database and import tools	Build the structure in Section 4 and the tool that reads the Excel export. Deliverable: a repeatable import.	1 to 2 weeks	
5. Portal build	Customer catalog and administration back office: search, drawings, hotspot editor, users, discounts, reporting, audit.	4 to 6 weeks	
6. FT3 Wagon data load	Import the export, upload the drawings, mark the callout positions, verify against the current portal.	2 to 3 weeks	Runs in parallel

7. Testing	Function, performance and security testing, plus RUF Diamond review against the agreed requirements.	1 to 2 weeks	
8. Launch and training	Release to selected customers alongside the existing portal. Train staff on the administration side and write the internal guide.	1 week	
9. Pilot period	The FT3 Wagon runs live while we measure how long a machine takes to maintain and gather customer feedback.	60 to 90 days	
Total to launch	From approval to the pilot machine being live	2 to 3 months	

The data load runs alongside the build rather than after it, since the drawings and the export can be prepared while the portal is being written. That overlap is a large part of why the timeline is short.

13.2 Why this is faster than a conventional estimate

- The design is already complete. On most projects this is the phase that consumes weeks of back and forth before anything can be built.
- Development is AI-assisted, which shortens the writing of the code itself considerably. It does not shorten the parts that need human judgement: agreeing requirements, checking the data is correct, and testing.
- The scope is one machine. The structure supports all three brands from the start, but only the FT3 Wagon is loaded for the pilot.
- The infrastructure is a single server with no complex integration. Xero comes later.

13.3 What could extend it

Two things realistically, and both are visible now rather than surprises:

- **The drawing files.** If the current portal will not release usable drawing images, they have to be produced from our own source documents, which adds time to the data load.
- **Marking the callout positions.** This is the one task that cannot be automated from the export. The FT3 Wagon has 44 drawings and 635 callouts. The pilot measures exactly how long this takes, which is what makes the remaining machines predictable.

13.4 Responsibilities

Area	RUF Diamond	Development
Requirements and design	Review and approve the design and this document.	Design provided. Adjust following review.
Accounts	Open the OVHcloud and GitHub accounts under a rufdiamond.com address.	Configure and secure the server.

Build	Answer questions on catalog structure and business rules.	Build the database, portal and import tools.
Data	Provide the drawing files. Verify the loaded machine is correct.	Import the data and mark the callouts.
Testing	Confirm it works for our own use.	Testing and corrections.
Launch	Select pilot customers and communicate the change.	Deploy, monitor, train and document.
After launch	Own all catalog content and customer administration.	Technical operation, patching, backups and monitoring.

13.5 What has to be ready before each stage

- Nothing can be built until the design and requirements are approved.
- The environment cannot be set up until the OVHcloud and GitHub accounts exist. Both can be opened now.
- The data load cannot start until the FT3 Wagon drawing files are supplied. The Excel export is already in hand.
- Launch cannot proceed until a restore from backup has been tested successfully.
- Retiring the current portal cannot be scheduled until its contract terms and notice period are known.

13.6 Recommended approach

- Run the FT3 Wagon as a controlled pilot before committing to the remaining machines.
- Keep the existing portal running throughout. Nothing is switched off until the replacement is proven.
- During the pilot, record enhancement ideas rather than building them, unless something is genuinely broken. A pilot with a moving specification does not finish.
- Load the IronHorse line first after the pilot, since it is not in the current catalog and adds coverage rather than duplicating it.

14. Recurring Costs

The figures below cover infrastructure and third-party services only. Development effort is not included here and is a separate matter.

14.1 Infrastructure, OVHcloud

Item	Cost	Notes
VPS-3 server	\$16.83 / month	6 vCPU, 12 GB RAM, 100 GB storage. Includes IPv4 address, unlimited bandwidth, anti-DDoS protection and standard daily backup. Published price.
Premium automated backup	From \$1.80 / month	Extends backup retention to seven rolling days and allows the backup time to be set. Recommended.

Snapshot option	From \$0.50 / month	Manual rollback point taken before significant updates.
Object storage	Approx. \$5 to \$15 / month	Diagram images and the independent nightly database copy. Depends on volume.
Additional storage	As required	Only if the catalog grows beyond the included 100 GB. Expandable to 5 TB.
Infrastructure subtotal	Approx. \$25 to \$40 / month	Roughly \$300 to \$500 per year.

14.2 Third-party services

Service	Cost	Notes
Domain name	Nil	A subdomain of rufdiamond.com, for example parts.rufdiamond.com. No registration cost.
SSL/TLS certificate	Nil	Let's Encrypt, free and automatically renewed.
GitHub	Nil	Private repositories and security scanning are free at our scale.
Email delivery service	Nil to approx. \$30 / month	For password resets, account notifications and system alerts. Free tiers cover several thousand messages a month, which is well beyond our expected volume. A paid tier applies only if volume grows.
Monitoring tools	Nil	Uptime Kuma, Prometheus and Grafana are open source and run on our own server. Hosted alternatives exist if preferred.
Database software	Nil	PostgreSQL carries no licence fee.
Operating system and web server	Nil	Ubuntu and Nginx are open source.
Application components	Nil expected	The standard components for a portal of this type are open source. If any paid component is proposed during development, it should be raised and approved before it is adopted.

14.3 Summary

Category	Per year	Confidence
OVHcloud infrastructure	\$300 to \$500	High. Published prices.
Third-party services	Nil to approx. \$360	High. Mostly free tiers.
Software licensing	Nil	High. All open source.
Total recurring	Approx. \$300 to \$860	Excludes development and technical support.

Recurring infrastructure and service costs are therefore modest. The significant costs in this project are development effort and ongoing technical support, both of which sit outside this section.

15. Required Inputs

- FT3 Wagon drawing files in the best available format, alongside the Excel export already provided.
- Confirmation of the data and drawings that can be exported from the current provider.
- Current annual cost, licence terms and notice period for the existing portal.
- Review and approval of the Figma design.
- Creation of the RUF Diamond GitHub organization and the OVHcloud account, both under a rufdiamond.com email address.
- Confirmation of who holds the technical support responsibility described in Section 10.

Recommended next step

Proceed with the FT3 Wagon as a controlled pilot. Approve the Figma design, open the OVHcloud and GitHub accounts, and obtain the FT3 Wagon drawing files so that the data migration can begin.